AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM240
Introduction to computer graphics

Final Examination
Spring Semester 2025/2026

Date: 6/6/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **100**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: This part consists of 10 multiple choice questions. For each question, encircle the label a, b, c, or d as correct answer. Each question carries 2 marks.

Question 1 [20 Marks]

1 The stroke of a line covers the attribute …….

A line width B color
C font D None of the above

2 The ……colour system is commonly used in printers.

A CIEXYZ B RGB
C CYMK D None of the above

3 An alpha-value is a number between 0.0 and 1.0 with ……..representing complete transparency .

A 0 B 10
C 0.5 D 1

4 The number of points for using the function QuadCurve2D.Float is ……

A 3 B 4
C 5 D 6

5 the junctions between individual Bezeier pieces in B-spline curve are called.

A modifiers B knots
C controllers D None of the above

6 The Java library …….contains advanced image processing and comprehensive image processing.

A AWT B Java2D
C Swing D JAI

7 The winding rule that require the directions of the lines defining a given shape is ……

A even-odd winding B non-zero winding
C Close-box winding D None of the above

8 the function …….. Is used to apply an operator on a given image.

A LookupOp B ConvolveOp
C filter D RescaleOp

9 In ……..there is no separation between modeling and rendering.

A Swing B AWT
C None of the above D None of the above

10 ……..is the most platform independent graphics programming type.

A Hardware- level programming B GKS
C Operating system-level programming D None of the above

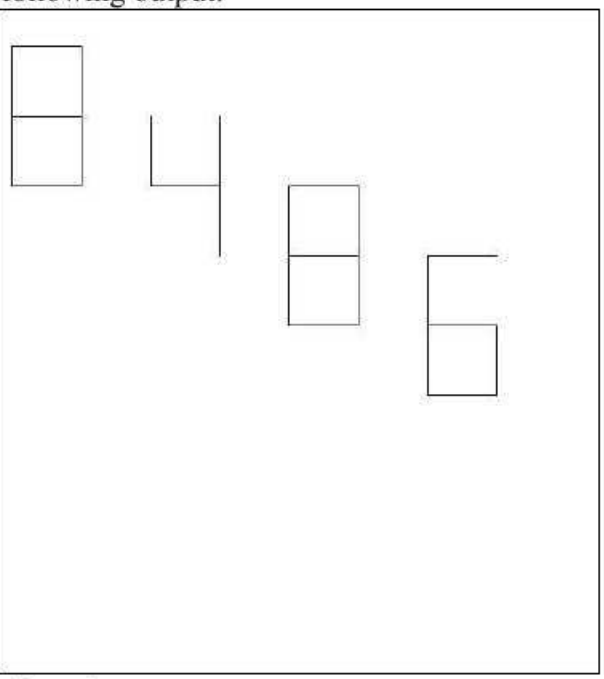
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Part II: This part consists of four questions with a total of 80 Marks, solve all questions.

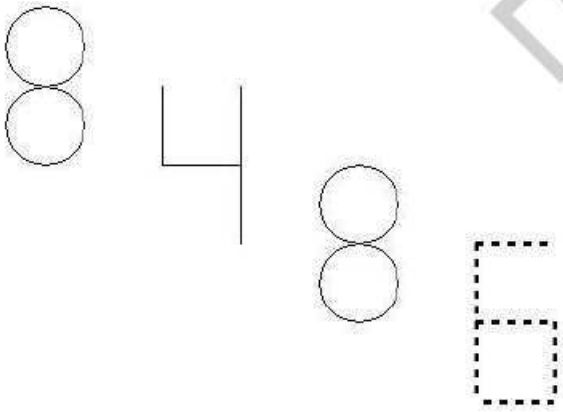
Question 2 [20 Marks]

Having the following Java code along with the following output:

```
g.drawLine(0,0,50,0);
g.drawLine(0,50,50,50);
g.drawLine(0,100,50,100);
g.drawLine(0,0,0,100);
g.drawLine(50,0,50,100);
g.drawLine(100,50,100,100);
g.drawLine(100,100,150,100);
g.drawLine(150,50,150,150);
g.drawLine(200,100,250,100);
g.drawLine(200,150,250,150);
g.drawLine(200,200,250,200);
g.drawLine(200,100,200,200);
g.drawLine(250,100,250,200);
g.drawLine(300,150,350,150);
g.drawLine(300,150,300,250);
g.drawLine(300,200,350,200);
g.drawLine(350,200,350,250);
g.drawLine(350,250,300,250);
```



Update the above code to output the following shapes:



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Question 3 [20 Marks- 5 marks per item]

a. Explain the different methods and constructors available for creating and setting colors in Graphics2D. Discuss the significance of the various color components, including RGB and alpha values.

b. Explain how RescaleOp modifies pixel values in an image. Include the formula and define what each parameter represents.

c. A kernel for ConvolveOp is defined as:

```
float[] kernel = {
    0, -1, 0,
   -1, 5, -1,
    0, -1, 0
};
```

I. What effect does this kernel have on an image?
II. How would you apply it in Java using ConvolveOp?

d. A student writes the following code to blur an image but it produces no visible effect:

```
float[] blur = {
    0, 0, 0,
    0, 1, 0,
    0, 0, 0
};
```

```
ConvolveOp op = new ConvolveOp(new Kernel(3, 3, blur));
BufferedImage result = op.filter(img, null);
```

I. Explain why the blur has no effect.
II. Suggest a correct kernel to apply a simple 3x3 averaging blur.

Question 4 [20 Marks]

Write a function in java code to implement Mandlbrot fractal sequence with maximum number of iterations of 2097152 for an arbitrary setting of the imaginary constant C.

Question 5 [20 marks]

a. Explain the role and components involved in the animation process of a graphical application. [5 marks]

b. Draft the output of this code:

```
g2.setComposite(AlphaComposite.getInstance(
    AlphaComposite.SRC_OVER, 0.6f));
g2.setColor(Color.BLUE);
Shape rectangle = new Rectangle2D.Double(50, 50, 200, 150);
g2.fill(rectangle);
g2.setComposite(AlphaComposite.getInstance(AlphaComposite.SRC_OVER, 0.5f));
g2.setColor(Color.RED);
Shape circle = new Ellipse2D.Double(150, 100, 150, 150);
g2.fill(circle);
g2.setComposite(AlphaComposite.getInstance(AlphaComposite.SRC_OVER, 1.0f));
g2.setColor(Color.BLACK);
g2.setStroke(new BasicStroke(2));
g2.draw(rectangle);
g2.draw(circle);
```

End of questions

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